

Rowan Callahan

Computational biologist • Seattle, WA

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Education

- **PhD, Biomedical Engineering**, Oregon Health & Science University *Expected fall 2026 (defended June 2026)*
 - Thesis: *Computational Methods for Resolving Contamination and Global Shifts in Cell-Free and Bulk RNA Sequencing*
- **BS, Bioengineering**, Cornell University *2014–2018*

Experience

- **PhD Researcher**, Center for Early Detection Advanced Research (Ngo Lab) *Fall 2020–2026*
 - Led work on GLORB, a method for differential expression testing using Bayesian inference in NumPyro/JAX.
 - Co-first author on Nature Communications paper demonstrating cancer prediction using cell-free RNA
 - Designed and implemented Snakemake pipelines processing terabytes of NGS data from hundreds of cell-free RNA samples
- **Research Assistant**, Maxson Lab, Oregon Health & Science University *January 2020–Fall 2020*
 - Created Snakemake pipeline for ATAC-seq data analysis processing hundreds of samples
 - Performed statistical analysis and data visualization using R/ggplot2 for leukemia progression studies; co-author on 3 published papers.
- **Laboratory Technician**, Brito Lab, Cornell University *Summer 2018–Summer 2019*
 - Developed Snakemake pipelines for genomic analysis of thousands of microbial genomes.
 - Conducted statistical analysis of microbiome datasets; co-author on 3 published papers.

Software

- **glorb-seq** — Bayesian RNA-seq normalization under global expression shifts (JAX/NumPyro).
pypi.org/project/glorb-seq • github.com/rowancallahan/GLORB-seq

Technical skills

- **Modeling**: Bayesian inference (JAX/NumPyro, Pyro), PyTorch, probabilistic graphical models, machine learning
- **Compute**: GPU training (NVIDIA/CUDA) with PyTorch and JAX; Slurm/HPC clusters
- **Bioinformatics**: RNA-seq, cell-free RNA, ATAC-seq, Snakemake pipelines, TCGA
- **Languages**: Python, R, Julia, bash
- **Tooling**: pandas, tidyverse; Linux administration, environment management

Selected publications

*Denotes equal contribution

Callahan, Rowan L., Stephen D. Coleman, and Thuy T. M. Ngo (Sept. 2026). “GLORB: Robust Bayesian inference for differential expression under global expression shifts”. In: *bioRxiv*. Preprint. DOI: [10.64898/2026.08.28.747928](https://doi.org/10.64898/2026.08.28.747928). URL: <https://doi.org/10.64898/2026.08.28.747928>.

Moore, Travis W*, **Spiliotopoulos, Elias***, **Callahan, Rowan L.***, C Ward Kirschbaum, Conner F Bailey, Hyun Ji Kim, Breeshey Roskams-Hieter, Florian Goncalves, Dove Keith, Aaron J Grossberg, et al. (2025). “Cell free RNA detection of pancreatic cancer in pre diagnostic high risk and symptomatic patients”. In: *Nature Communications* 16.1, p. 7345.

Roskams-Hieter, Breeshey, Hyun Ji Kim, Pavana Anur, Josiah T Wagner, **Rowan L. Callahan**, Elias Spiliotopoulos, Charles Ward Kirschbaum, Fehmi Civitci, Paul T Spellman, Reid F Thompson, et al. (2022). “Plasma cell-free RNA profiling distinguishes cancers from pre-malignant conditions in solid and hematologic malignancies”. In: *NPJ precision oncology* 6.1, p. 28.

Leadership and service

- **External Vice President**, Biomedical Engineering Society, OHSU *2023–2024*
- **Mentor**, Undergraduate Summer Research Program, OHSU *Summer 2024*